

Fretology

Gaining luthier perspectives toward informing project development



By

Haoyan Zhong

Weilun Deng

Botao Hu

Submission Date

August 30, 2024

A Worcester Polytechnic Institute Interactive Qualifying Project

Submitted To

Professor V.J. Manzo

The Electric Guitar Innovation Lab (EGIL)

WORCESTER POLYTECHNIC INSTITUTE

100 Institute Rd, Worcester, MA 01609

Abstract

Fretology is an acoustic guitar research, preservation, and education project. To clarify the future direction of Fretology, the project investigated the attitudes and thoughts of luthiers towards Fretology through two main methods. Questionnaires were used to ascertain the basic profile of the luthiers and their interest in the project, while interviews were conducted to discuss in-depth attitudes towards Fretology and its future direction and possibilities. The team concluded that Fretology still has great potential and plenty of room for development.

Executive Summary

This report presents the findings of an Interactive Qualifying Project aimed at enhancing Fretology, an acoustic guitar research, preservation, and education project. Fretology, developed by the Electric Guitar Innovation Lab (EGIL) at Worcester Polytechnic Institute (WPI) in collaboration with the Quinnipiac Radiologic Sciences program, seeks to make acoustic and physical data of historically significant guitars accessible to luthiers, musicians, educators, and researchers. The project currently hosts eight 3D models of guitars on its website, serving as a valuable resource for its target audience.

Despite the significant progress made by the founders and previous research teams, Fretology still faces challenges in engaging its target audience and addressing the specific needs of luthiers and researchers. Our team, consisting of three undergraduate students and a faculty advisor, wanted to address these challenges by focusing on two primary objectives: attracting more related groups and individuals to Fretology and identifying their important needs.

The research was conducted using a mixed-methods approach, including both surveys and interviews. The surveys were distributed to several online luthier communities to get quantitative data on luthiers' involvement in related events and activities, while the interviews focused on gathering professional opinions and qualitative results from experienced luthiers on the Fretology project.

The results obtained from the survey and interviews show the potential of Fretology in the craft of luthiery, especially for younger luthiers who are more open to modern technologies like 3D modeling and CT scanning. However, the project still faces several limitations, including the

need for more detailed and accurate models, better integration of 3D models with traditional 2D drawings, and collaboration with large corporations or individual collectors in aid of the preservation of rare or valuable guitars.

Table of Contents

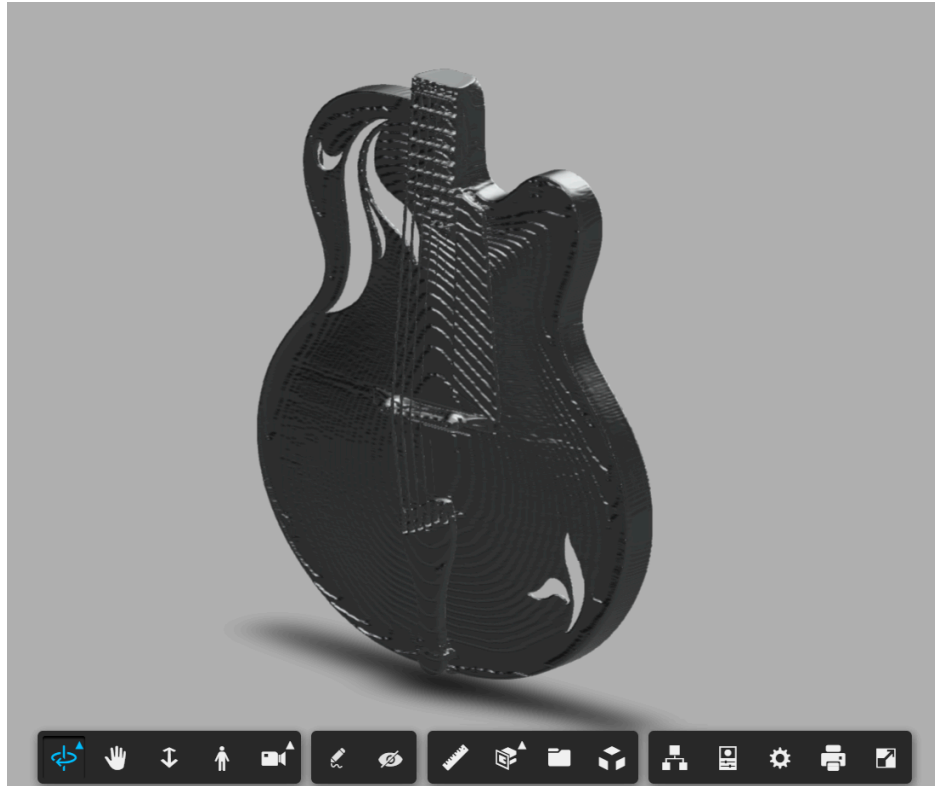
Abstract.....	1
Executive Summary.....	2
Table of Contents.....	4
Introduction & Background.....	5
Methodology.....	7
Findings.....	13
Conclusion and Recommendations.....	23
Appendix A: Survey Details.....	25
Appendix B: IRB Application.....	29

Introduction & Background

The acoustic guitar holds a significant place in musical history, influencing many different music genres and cultures around the world. As the appreciation for acoustic guitars continues to grow, the importance of preserving and studying these iconic instruments has also risen over time.

According to the official website of the Fretology project, “Fretology is an acoustic guitar research, preservation, and education project”. The primary goal of Fretology is to make both acoustic and physical data of historically significant guitars accessible to luthiers, musicians, educators, and researchers. By doing so, it serves as a useful resource to its target audience, allowing for a deeper understanding and appreciation of these instruments.

The Electric Guitar Innovation Lab (EGIL) at Worcester Polytechnic Institute (WPI), in collaboration with the Quinnipiac Radiologic Sciences program and many other researchers, has been developing and maintaining the Fretology project. With the great amount of effort made by them, a complete website is currently presented with eight 3D guitar models.



A Benedetto Blue 3D model on the Fretology website.

Although a huge foundation was made by the EGIL and previous research teams, the Fretology project still faces challenges. One most important challenge is to grow focus and engagement among its target audience, and additionally, to identify and address the specific needs of luthiers and researchers from Fretology.

The IQP team is composed of three undergraduate students and a faculty advisor. Given the challenges upfront, the team focused on two primary objectives: attracting more related groups and individuals to Fretology, and identifying the important needs from them. With the limited team size, direct improvements to the Fretology project can be limited, however, these results can be useful in future research and developments.

Methodology

Overview

The goal of our project was to find out how Fretology can be improved. To achieve this goal,

Our team used the method that includes two main steps:

1. Survey

2. Interview

By that information our team could answer where the luthiers are and identify what strengths and weaknesses Fretology has and what's the future direction of Fretology from a valid perspective.

Survey

Our target group for this survey is luthiers, with the aim of gathering comprehensive quantitative data to better understand their online activities, resource usage, and opinions on the Fretology project. Our team set a target sample size of 50 luthiers and created a questionnaire using Qualtrics. A complete list of the survey questions can be found in **Appendix A**. The survey was distributed through various online platforms, including the Luthiers Subreddit, professional forums, and social media groups belonging to luthiery.

The survey was designed to explore key areas such as the platforms luthiers frequent, how they interact with peers, and their familiarity with the resources provided by Fretology. It consisted of

multiple-choice and open-ended questions aimed at understanding the accessibility and activity levels of these luthiers, their familiarity with Fretology, and their opinions on its current offerings.

With this survey, our team aimed to learn more about two key topics: where luthiers are active and where they socialize. The Luthiers Subreddit was chosen for its wide accessibility and large number of active luthiers; however, due to limited advertising, the initial data collection was restricted. Despite this, the team found success in more professional luthier websites, such as The Acoustic Guitar Forum and the Official Luthiers Forum, which allowed for sufficient data collection and led to a large number of luthiers agreeing to participate in subsequent interviews.

Key questions in the survey included:

- **Professional Skill Level:** Luthiers were asked to self-assess their skill level on a scale from novice to expert, providing insight into the variation in craftsmanship among respondents. This question is crucial for understanding how different skill levels might influence their opinions on supplementary tools like CT scanning during the guitar-making process. According to our projections, the demand for Fretology may decrease as luthiers' skill levels increase.
- **Online Communities:** The team sought to identify the most active online communities where luthiers engage, helping the team pinpoint the platforms most relevant to them.
- **Event Participation:** The team explored which in-person events luthiers attend to understand their professional networks and community involvement.

- **Relevant Journals:** Respondents were asked about the luthier-related journals they read, helping us determine their sources of professional knowledge and trends.
- **Social Platforms:** Our team inquired about the social media platforms luthiers use to connect with peers, offering insights into their communication habits. This question was also instrumental in identifying those luthiers who might be interested in participating in further interviews.
- **Technology Acceptance:** Another crucial question asked respondents to what extent CT scan images showing the internal structure of guitars might inform their work. This question was designed to link the survey to the Fretology project and to gauge respondents' attitudes toward this new technology, as their perspectives on this matter would likely reflect their views on Fretology.

Interview

In addition to conducting surveys, the team also carried out detailed interviews to gather qualitative insights, which allowed the team to delve deeper into the professional perspectives of experienced luthiers. These interviews aimed to provide a more nuanced understanding of their needs, expectations, and opinions on the Fretology project, especially regarding the practical application of new technologies like CT scanning and 3D modeling.

Our team planned to interview approximately 5-10 luthiers. This sample size was chosen to ensure a diverse range of perspectives while remaining manageable within our time and resource

constraints. By selecting luthiers with varying levels of expertise and backgrounds, the team aimed to gather comprehensive and rich data that reflects the diversity of the luthier community. The chosen sample size also allowed for each interview to be conducted in-depth and thoroughly, without compromising quality due to time or resource limitations.

Interview Process:

1. **Introduction:** Each interview began with an introduction to our project and an explanation of Fretology's purpose. This included an overview of our current work, the resources available on the Fretology website, and the specific goals of our research. Our team provided context on how Fretology aims to assist luthiers by making both acoustic and physical data of historically significant guitars accessible to them.
2. **Discussion:** Following the introduction, the team structured the discussion around several critical topics to gain detailed insights:
 - **Current State:** Our team asked luthiers for their opinions on the current state of Fretology, including the usability of the website and the quality of the resources available. This helped the team understand what aspects of the project are working well and what areas might need improvement.
 - **Strengths and Weaknesses:** Luthiers provided their insights into what Fretology is doing well and where it could improve. This included feedback on specific features, the accuracy of 3D models, and the relevance of the available data to their work.

- **Future Directions:** Our team sought suggestions for new features, content, or collaborations that could enhance the value of Fretology for the luthier community. Luthiers were encouraged to think about how Fretology could evolve to better meet their needs in the future.
- **Feedback and Suggestions:** Our team maintained an open-ended format to capture a wide range of perspectives, encouraging interviewees to provide any additional feedback or ideas they had. This approach allowed the team to gather insights that might not have been covered by the structured questions.

Key Areas of Focus:

During the interviews, Our team aimed to explore the following aspects:

- **Technology Acceptance:** Our team assessed the luthiers' understanding and acceptance of CT scanning and 3D modeling technologies. This included gathering feedback on their experiences with these technologies in their actual operations and how they perceive their utility in luthiery.
- **Practical Application:** Our team investigated the specific applications of these technologies in luthiery. Luthiers shared how they currently use or could potentially use the resources similar to Fretology in their work, particularly the 3D models and CT scans.
- **Demand Analysis:** Our team analyzed the specific needs and demands of luthiers, focusing on what resources or features would be most beneficial to them.

This helped the team identify gaps in Fretology's offerings, opportunities for development, and future direction.

- **Suggestions for Improvement:** Luthiers were asked to provide suggestions for improving Fretology. This included recommendations on enhancing the accuracy of the models, adding new features, and expanding the range of guitars and data available through the platform.

By focusing on these aspects, the interviews provided our team with a deeper understanding of how Fretology can better serve the luthier community and guide the future development of the project. The insights gained from these interviews are invaluable for evaluating the practical utility of Fretology's resources and setting a clear direction for future improvements.

Findings

Introduction

Within 30 days of the first distribution of the survey, a total of 49 responses were collected. To ensure the relevance of the data, the team filtered out incomplete or non-significant responses. As a result, 28 responses were in the end used for data analysis in the following section.

In addition to the survey, the team wanted to deepen their understanding through follow-up interviews with respondents. Although 7 luthiers have expressed their interest in participating in an interview, the team faced time ability and communication challenges with the luthiers, which restrained the team's ability to successfully schedule and conduct interviews. In the end, the team was only able to interview one of the luthiers. However, by successfully connecting to a local luthier group in China, the team was able to conduct interviews with three additional luthiers. These 4 interviews were extremely valuable in understanding different luthiers' opinions on the Fretology project.

Results and findings from both the survey and these interviews will be discussed separately in detail in the following sections.

Survey

As previously mentioned, the survey mainly serves to get a deeper understanding of what online resources luthiers use and what events they often attend. The survey responses came from luthiers with varying levels of profession. The analysis of each question is provided below.

The first question asked respondents to self-assess their skill as luthiers on a scale from 1 to 5, with 1 representing Novice and 5 representing Expert. The average rating was 3.53, which is above the median score of 3, indicating the relatively high professional background of the respondents.

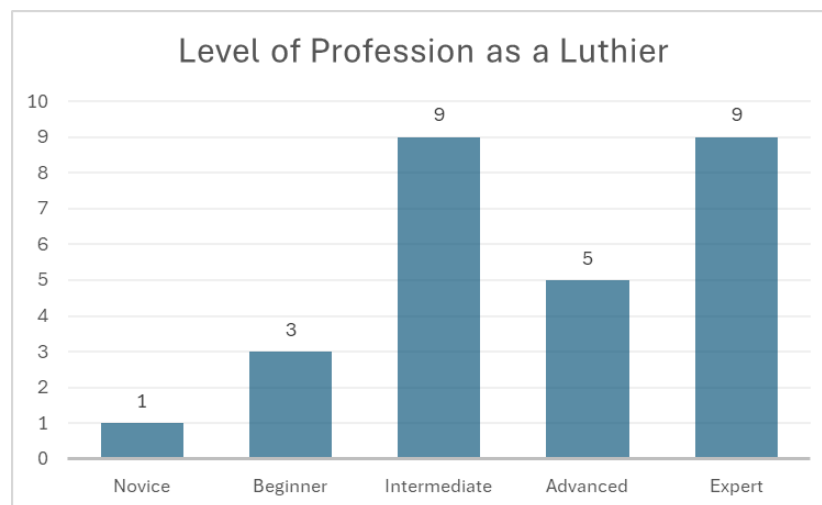


Figure 1: Chart of respondents' self-assessing levels of profession as a luthier.

The second question aimed to identify the most active online luthier communities. According to the data, in addition to the forums where the survey was distributed—The Official Luthiers Forum, Acoustic Guitar Forum, and Luthiers Subreddit—two mostly mentioned forums are the Musical Instrument Makers Forum (MIMF) and the Delcamp Classical Guitar Forum. Several

other communities are also mentioned by multiple respondents, including the Luthiers Club Facebook Group, The Unofficial Martin Guitar Forum, and The Looth Group.

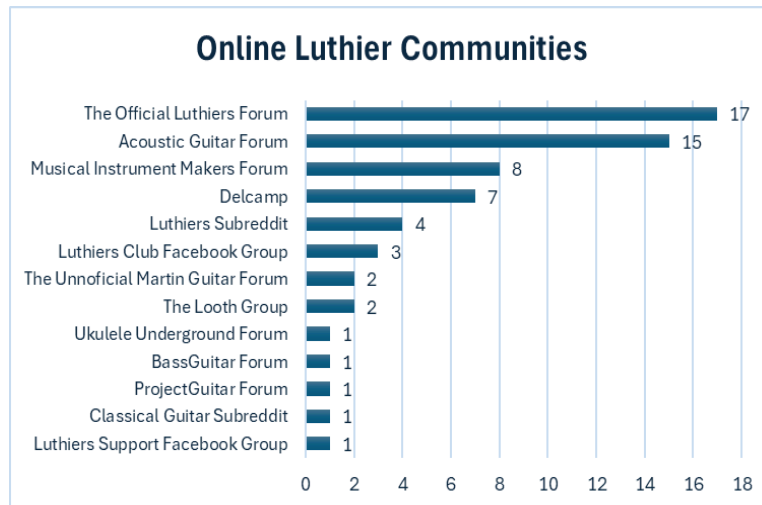


Figure 2: Chart showing the times of each online luthier community mentioned by respondents.

Next, the third question explored the in person events that luthiers attend, and answers to this question varies a lot. While 7 respondents indicated that they had not attended any luthier-related events, 8 out of the remaining 21 respondents mentioned they had attended the Guild of American Luthier Convention, and 6 referred to the Woodstock Invitation Luthier Showcase. Additionally, the ASIA Symposium, Healdsburg Guitar Festival, and Fretboard Journal Summit each received 4 mentions.

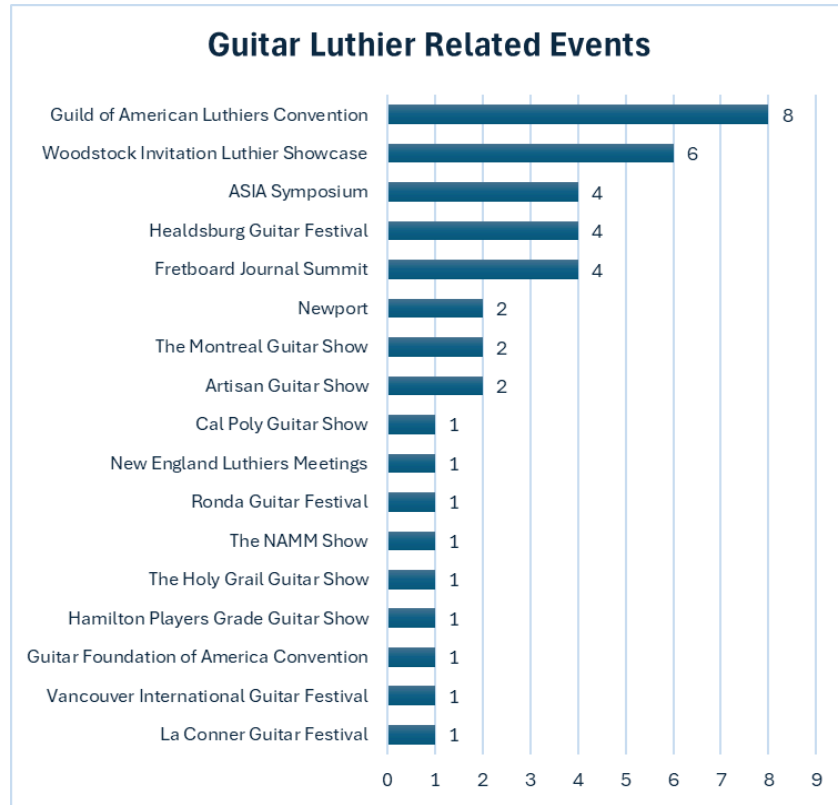


Figure 3: Chart showing the times of each guitar luthier-related event mentioned by respondents.

The fourth question inquired about the luthier-related journals respondents read. According to the responses, 17 respondents reported that they have read the Guild of American Luthiers Journal, 6 mentioned the Fretboard Journal, while Guitarmaker by ASIA was noted by 5 respondents, and 2 respondents have also mentioned the Catgut Acoustical Society Journal.

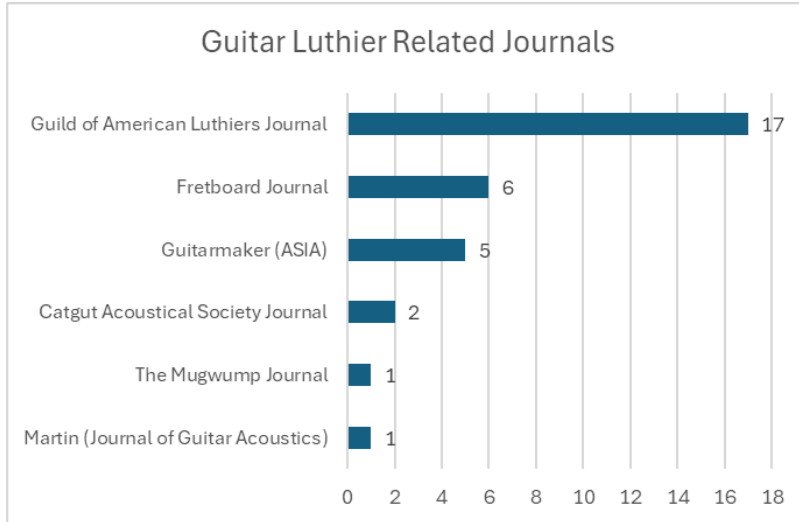


Figure 4: Chart showing the times of each luthier-related journal mentioned by respondents.

The fifth question focused on the social platforms respondents use to connect with fellow luthiers. The data suggests that Facebook and Instagram are the most commonly used platforms, though a few respondents mentioned using Discord and Yahoo.

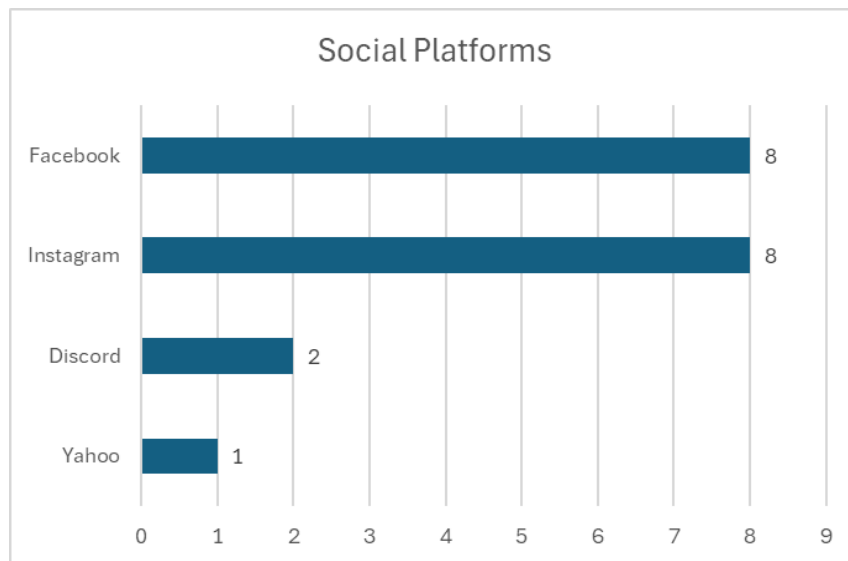


Figure 5: Chart showing the times of each social platform mentioned by respondents.

Finally, the last two questions aimed to assess luthiers' acceptance to modern technology. Specifically, question six asked for their opinions on the usefulness of CT scanning for showing internal structure of guitars, while question seven asked about the frequency of them implementing new tools in their instrument making process. As reflected in the charts, respondents generally have an above average attitude toward CT scanning technology, though many indicated that they do not implement new tools on their own.

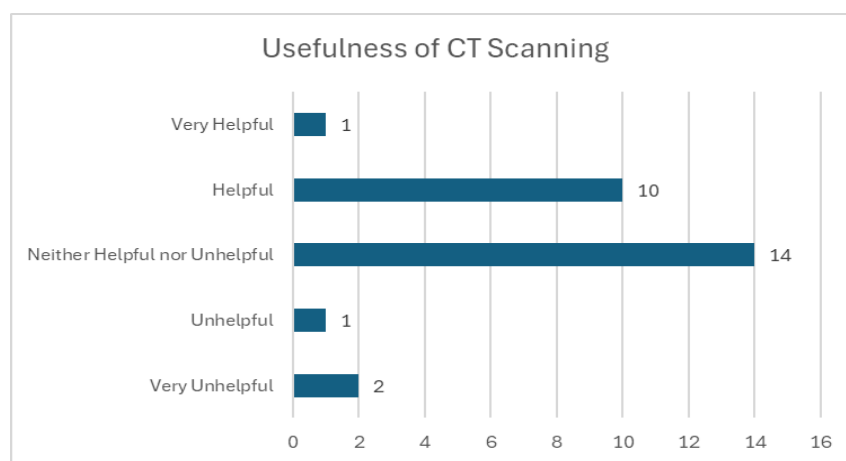


Figure 6: Chart showing responses on the usefulness of CT scanning.

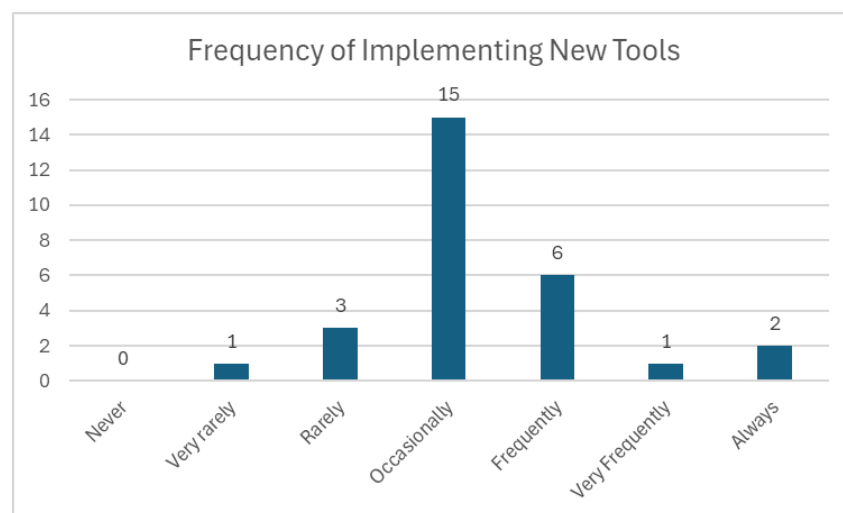


Figure 7: Chart showing responses on the frequency of respondents implementing new tools

Interview

From our research, the team was able to develop a clear picture on the future improvement of Fretology. With a series of interviews, surveys, and conversations with multiple expert luthiers all over the world. This work led to the following results from the project:

- The Potential of Fretology
- The Improvement of Fretology Website
- The Protection of Old Guitars
- The Help to the Luthiers

To explore these findings more in-depth, this chapter will focus on the specific learned from the interviews which were speaking with the different expert luthiers and summaries the team made to further our project. The above results will be expanded and summarized. All quotations were from our interviews.

The Potential of Fretology

Younger luthiers are more accepting of the idea of Fretology and think it has significant development potential, in contrast to a few more established luthiers who think it has less room to grow.

--I believe that we are currently seeing a new bottleneck in guitar making. However, I don't believe the models shown in the website have enough potential to enable guitar manufacturing to

advance beyond its current state. Since it isn't suggestive or precise enough to be of any service to the luthiers. So the potential for development depends on huge advances in technology.”

Numerous luthiers with pertinent knowledge have conveyed the benefits of this technique and the enormous commercial potential it has.

“--After browsing the website, do you have any thoughts on it in general?

--I think the potential and significance of this project lies in the fact that if it succeeds, it will be a further step forward in the craft of luthiery. As I know, plenty of big companies are working in this direction, so there may be room for cooperation with them in the future.”

The Improvement of Fretology Website

The luthier's assessment of the 3D model and this technology is likewise divided, although overall results indicate that it is well developed. Some younger luthiers believe that the accuracy is not up to par, particularly when it comes to the distance display, body surface, body curvature, bridge details, sound beam accuracy, and other details that are hard to depict. On the other hand, some luthiers believe that the 3D model that is currently on display is adequate and that hand drawing is necessary to improve the relevant accuracy.

“-With the help of hand drawing, the main improvement in this area is that I can easily calculate the air volume inside the guitar case from the updated 3d model, which is a huge step in sound reproduction.”

The luthiers offered a number of recommendations on the necessity for more details. Images of the original guitar and recordings of its sound have been referenced several times. The reason is to repair a guitar, the level of sound restoration is unquestionably the most important factor. The 3D model alone is only able to depict the general form or overall appearance of the finished product; it is unable to comprehend the production process itself. The importance of this 3D model will be substantially increased by the supplement of details on the composition and materials of each guitar component.

The Protection of Old Guitars

Although there is still much to be done in this area, a great deal more information, pictures, and drawings will be required in order to assist future generations in trying to recreate the exact form and sound of a guitar when they get the chance.

“--With the advancement of technology, you should be able to work with huge corporations or individual collectors, which will be crucial for the preservation of rare or valuable guitars.”

The Help to the Luthiers

“I do think it can help luthiers during the making process, but for us luthiers who have been making guitars for a long time it's minimal because we don't use this kind of technology for our year-round work or hobby, and we can say that using traditional 2D drawings as well as

traditional craftsmanship is our comfort zone, and it's too hard for us to change or be exposed to new technology. ”

Because they have a comfort zone and some of them don't even recognize the technology, senior luthiers are undoubtedly less willing to adopt new technology than younger luthiers. The development and availability of Fretology is currently of minimal assistance to luthiers; however, as more information becomes available and more guitars are added to the system, it may be possible to construct a comprehensive system. Younger luthiers are far more accepting of new technology and recognize its potential based on their experience with it.

Conclusion & Recommendations

The study primarily looks into Fretology's development potential and trend, as well as luthiers' attitudes regarding it. The majority of the luthiers, it turned out, were enthusiastic about the project and offered a lot of recommendations for its future course. To improve the project website, we need to make more materials accessible, integrate 3D models into 2D drawings, and create dynamic models to improve the role of luthiers as guides. Going forward, Fretology should collaborate with major corporations or independent luthiers to enhance model accuracy, expand the model count, and finally promote itself.

Many luthiers have mentioned the need to provide more material for their models after reading the website. Here are three categories into which this may be divided: 1. The substance of every component of the guitar, including the matching length, thickness, density, etc. 2. Images of the original guitar taken from various perspectives, particularly the beam, which improve the model's overall correctness and aid the luthier in comprehending the guitar's construction. 3. The original guitar's tone, which is crucial for some luthiers who wish to reappear those great guitars because a nice guitar's tone is one of the standards.

Fretology's instruction for luthiers will be much improved if it can convert the 3D model to 2D drawing. The ability to make 2D drawings from various angles of a 3D model would be a clear and useful reference for research or restoration work, as most luthiers still design their guitars using 2D drawings.

The dynamic model, which can display the amplitude of the neck following string variations, the air flow rate, and the sound generated, offers a vast amount of potential for future growth.

Overall, Fretology shows strong potential at the present, but additional material needs to be added in many areas to support its ability in helping and guiding luthiers.

Appendix A: Survey Details

Thank you for taking the time to participate in our survey!

This survey is part of Fretology (fretology.org), an acoustic guitar research, preservation, and education project. Fretology is dedicated to the in-depth analysis of exceptionally-built, historically significant guitars, and is developed and maintained by the Electric Guitar Innovation Lab (EGIL) of WPI.

Although Fretology is still in its early stages, we are committed to making our data open and accessible to luthiers, musicians, educators, and researchers. We believe in the power of collaboration and welcome partnerships as we continue our work. Your insights and feedback are invaluable to us as we strive to enhance the impact of Fretology.

Please be assured that your responses will remain confidential and will not be disclosed to any third parties.

Q1. Regarding your skills as a luthier, at what level of professional would you rate yourself?

- ☐ Novice
- ☐ Beginner
- ☐ Intermediate
- ☐ Advanced
- ☐ Expert

Q2. In which of the following forums or online communities have you participated? (Select all that apply)

- ☐ The Official Luthiers Forum
- ☐ Acoustic Guitar Forum
- ☐ Luthiers Club Facebook Group
- ☐ The Musical Instrument Makers Forum (MIMF)
- ☐ Luthiers Subreddit
- ☐ Others _____
- ☐ None

Q3. From which of the following conferences or events have you attended? (Select all that apply)

- ☐ American Guild of Luthiers Convention
- ☐ Fretboard Journal Summit
- ☐ Woodstock Invitation Luthier Showcase
- ☐ Guitar Foundation of America Convention (11)
- ☐ La Conner Guitar Festival (5)
- ☐ Artisan Guitar Show (8)
- ☐ Vancouver International Guitar Festival (3)
- ☐ Others (4) _____
- ☐ None (13)

Q4. Which of the following journals have you read? (Select all that apply)

- ☐ Guild of American Luthiers Journal
- ☐ Fretboard Journal

- ☐ Others _____
- ☐ None

Q5. From the following social platforms, select the ones you use to build connections with other luthiers. (Select all that apply)

- ☐ Facebook
- ☐ Discord
- ☐ X
- ☐ Instagram
- ☐ Telegram
- ☐ Snapchat
- ☐ Slack
- ☐ Others _____

Q6. To what extent might CT scan images showing the internal structure of guitars inform your work?

- ☐ Very helpful
- ☐ Helpful
- ☐ Neither helpful nor unhelpful
- ☐ Unhelpful
- ☐ Very unhelpful

Q7. If you would like to participate in a follow-up interview or questionnaire to leave your feedback on this project, please enter your email below.

Q8.To help us improve the survey quality, please leave any comments or suggestions on the questionnaire you are taking.

Q9.Please leave any additional information that you want us to know.

Appendix B : IRB Application

This appendix section consists of all materials used in the IRB application, including the advertising material, initial methodology draft, and informed consent form.

Advertising Material

We are WPI Class of 2026 students working on an IQP project called "Fretology." We want to use this survey to know where the luthiers get their information and where they socialize.

This survey's anonymous methodology ensures that none of your personal information is disclosed. Your participation and feedback will be very valuable to our research, we greatly appreciate your feedback and ideas.

The following is the link to the survey: (link to the survey)

Initial Methodology Draft

The method used in this project involves a two-step approach. The first step involves conducting a survey targeting a broad range of luthiers, while the second step includes follow-up interviews with a focus group interested in the Fretology project.

The goal of the first step is to understand which sources of information luthiers use and where they socialize. To achieve this, a survey is conducted, starting from some of the major online

communities. Once initial responses are obtained, the survey can be further spread to other communities.

The second step involves follow-up interviews to collect feedback on the Fretology project, focusing on its relevance to luthiers and identifying potential improvements. In the interviews, participants are shown the Fretology website and its key features. They are asked to share their thoughts and suggestions regarding the website and the overall project.

Informed Consent Form

Investigator: V Manzo, Haoyan Zhong, Botao Hu, Weilun Deng

Contact Information: vjmanzo@wpi.edu , hzhong@wpi.edu, bhu3@wpi.edu, wdeng1@wpi.edu

Title of Research Study: Fretology- an open guitar research, preservation and education resource IQP E term

Sponsor: N/A

Introduction: You are being asked to participate in a research study. Before you agree, however, you must be fully informed about the purpose of the study, the procedures to be followed, and any benefits, risks or discomfort that you may experience as a result of your participation. This form presents information about the study so that you may make a fully informed decision regarding your participation.

Purpose of the study: We want to investigate luthier's knowledge of Fretology and whether the development of Fretology could help them.

Procedures to be followed: You will be first asked to answer a series of questions that are related to your experience and understanding as a luthier from a survey. At the end of the survey, you can choose to participate in an interview, in which you will be asked for your thoughts and opinions on the Fretology project and its website.

Risks to study participants: No risk greater than experienced in everyday life

Benefits to research participants and others: An improvement in Fretology can create a great tool/study for the luthiers/guitar musicians to make excellent guitars.

Record keeping and confidentiality: We will store the relevant information we collect and interview information in Qualtrics and Google doc. Access to and editing of the information is vested only in the investigators mentioned above. Records of your participation in this study will be held confidential so far as permitted by law.

However, the study investigators, the sponsor or its designee and, under certain circumstances, the Worcester Polytechnic Institute Institutional Review Board (WPI IRB) will be able to inspect and have access to confidential data that identify you by name. Any publication or presentation of the data will not identify you.

For more information about this research or about the rights of research participants, or in case of research-related injury, contact:

V Manzo, vjmanzo@wpi.edu, Faculty Advisor

Haoyan Zhong, hzhong@wpi.edu, Student Investigator

Weilun Deng, wdeng1@wpi.edu, Student Investigator

Botao Hu, bhu3@wpi.edu, Student Investigator

Ruth McKeogh, Tel. 508-831-6699, irb@wpi.edu, IRB Manager

Gabriel Johnson, Tel. 508-83-4989, gjohnson@wpi.edu, Human Protection Administrator

Your participation in this research is voluntary. Your refusal to participate will not result in any penalty to you or any loss of benefits to which you may otherwise be entitled. You may decide to stop participating in the research at any time without penalty or loss of other benefits. The project investigators retain the right to cancel or postpone the experimental procedures at any time they see fit.

By signing below, you acknowledge that you have been informed about and consent to be a participant in the study described above. Make sure that your questions are answered to your satisfaction before signing. You are entitled to retain a copy of this consent agreement.

Study Participant Signature

Date: _____

Study Participant Name (Please print)

Signature of Person who explained this study Date: _____
